SENTINELX V7 ULTIMATE

Autonomous Public Safety AI Architecture & Complete Technical Manual

Hackathon / Problem Statement:	Economic Times AI Hackathon 2026 — Problem Statement 6: AI for Digital Public Safety
System Classification:	National Security & Citizen Protection Autonomous Multi-Agent Defense Grid
Author / Maintainer:	MEETDesai011 (meetdesai0311@gmail.com)
Model Accuracy Achieved:	95.06% Multi-Spectral Calibrated Ensemble on 344 Validation Banknotes
Release Artifacts Included:	Web Command Centre (Vite), Mobile App (React Native APK), 7 Microservices, 17-Agent Mesh

1. Executive Overview & Core Problem Statement

SentinelX V7 Ultimate is an autonomous, real-time national digital public safety defense grid engineered to protect Indian citizens and empower Law Enforcement Agencies (LEAs) against the modern digital fraud crisis. It addresses four highly destructive crime vectors currently facing Indian law enforcement and financial systems:

- **Digital Arrest Extortion:** Syndicates impersonating CBI, ED, Cyber Crime, and Supreme Court officials over Skype/WhatsApp video calls, coercing victims into liquidating their lifetime savings.
- **High-Fidelity Counterfeit Banknotes:** Sophisticated counterfeit Indian currency circulating across ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, and ₹2000 Mahatma Gandhi series denominations.
- **Multi-Hop Mule Account Networks:** Rapid layer-by-layer UPI and IMPS transactions siphoning extorted funds through layered accounts within seconds to defeat traditional manual police tracing.
- **Court-Admissible Digital Evidence:** Lack of immediate, cryptographically verifiable forensic evidence packages compliant with Section 65B of the Indian Evidence Act.

2. System Architecture & High-Level Data Flow

SentinelX is built as a cloud-native, microservices-driven monorepo integrating a React Web Command Centre, a React Native Citizen Mobile Application, seven high-throughput FastAPI/Express backend microservices, and an autonomous 17-Agent Mesh consensus network. Below is the operational data flow:

- **Step 1 — Ingestion & Telemetry Capture:** Citizen Mobile app or telecom gateways forward voice/audio transcripts, SMS payloads, banking transfer data, or banknote scan photographs to the API Gateway.
- **Step 2 — Specialized AI Microservice Routing:** API Gateway dispatches payloads to specialized microservices: speech/text to Scam Detection, banknote scan to Counterfeit Vision, transaction pairs to Fraud Graph, and suspect entities to Threat Fusion.
- **Step 3 — 17-Agent Autonomous Mesh Consensus:** Decentralized AI agents (investigator, forensics, legal, banking liaison, enforcement) correlate multi-modal findings and execute automated remediation protocols.

- **Step 4 — Real-Time LEA Command Broadcast:** High-severity alerts are pushed via WebSockets to the Command Centre UI, providing geo-spatial heatmaps, mule account graph topology, and real-time intervention options.
- **Step 5 — Autonomous Forensic Evidence Packaging:** An automated PDF report sealed with a SHA-256 cryptographic hash and Section 65B compliance certificate is generated and stored for court prosecution.

3. Complete Code File Directory & Exact Job Responsibilities

Every single file in the SentinelX codebase has a dedicated, production-grade responsibility. Below is the complete file-by-file breakdown explaining the exact job, implementation technologies, and interfaces:

Root Configuration & Infrastructure

Code File Path	Exact Job & Technical Responsibility
docker-compose.yml	Orchestrates the entire infrastructure: PostgreSQL 15, Neo4j 5.12, Redis 7, Apache Kafka (KRaft), API Gateway, and 6 FastAPI microservices on isolated networks with health checks.
package.json	Root monorepo configuration managing npm workspaces, build orchestration, cross-app compilation scripts, and repository dependencies.
.env.example	Template for all environment secrets: database connection URIs, JWT signing secrets, Neo4j credentials, Kafka broker addresses, and service endpoints.
.gitignore	Excludes build artifacts, node_modules, Android Gradle output, temporary cache files, and intermediate model extraction npy caches.

17-Agent Mesh (agents/)

Code File Path	Exact Job & Technical Responsibility
agents/agent_mesh.py	Implements the autonomous 17-agent consensus network. Defines specialized agent roles (Investigator, Telecom Analyst, Graph Forensics, Mule Tracker, Legal Compliance, Enforcement). Handles peer-to-peer message passing, distributed consensus voting, and automated remediation actions.
agents/test_agent_mesh.py	Automated test suite validating agent communication, voting thresholds, consensus resolution, and incident escalation protocols.
agents/requirements.txt	Defines Python runtime dependencies for the agent framework: pydantic, asyncio, websockets, and httpx.

Command Centre Web Dashboard (apps/command-centre/)

Code File Path	Exact Job & Technical Responsibility
apps/command-centre/src/App.tsx	The flagship Law Enforcement Agency (LEA) Command Centre single-page application. Features interactive GIS threat heatmaps (Leaflet), Digital Arrest Incident Simulator with real-time call transcript evaluation, Counterfeit Vision pipeline with client-side canvas pre-cropping and chromatic separation, Mule Account Graph explorer, Citizen Inbox, Evidence Package manager, and live Agent Mesh telemetry monitor.
apps/command-centre/src/main.tsx	Application root bootstrapping the React 18 DOM tree, strict mode boundaries, and

	global styles.
apps/command-centre/vite.config.ts	Vite build configuration with React plugin, chunk splitting, and rapid hot module replacement (HMR).
apps/command-centre/tailwind.config.js	Custom design system definitions: cyber-defense dark palette, glowing neon status indicators, glassmorphic card utilities, and responsive breakpoints.
apps/command-centre/package.json	Defines frontend dependencies: React 18, Lucide React icons, Leaflet GIS, Tailwind CSS, TypeScript 5, and Vite 5.

Citizen Mobile Application (apps/citizen-mobile/)

Code File Path	Exact Job & Technical Responsibility
apps/citizen-mobile/App.tsx	Production React Native citizen security heads-up display (HUD). Provides real-time call scanner monitoring for suspected digital arrest patterns, camera-based physical banknote counterfeit scanner, panic SOS trigger, and instant crime reporting.
apps/citizen-mobile/app.json	Expo configuration declaring app metadata, dark mode system styling, Android package ID ('com.sentinelx.citizenmobile'), and iOS bundle ID.
apps/citizen-mobile/package.json	Dependencies: React Native 0.74.5, Expo SDK 51, TypeScript, and native mobile UI primitives.
apps/citizen-mobile/android/	Full native Android Gradle project containing gradlew scripts, AndroidManifest.xml, build.gradle, and resource bundles for compilation into standalone Android APK.

API Gateway Service (services/api-gateway/)

Code File Path	Exact Job & Technical Responsibility
services/api-gateway/src/index.ts	Central Express & TypeScript API Gateway. Handles JWT authentication, RBAC authorization, request rate limiting, reverse proxying to microservices, and PostgreSQL persistence via Prisma.
services/api-gateway/prisma/schema.prisma	Relational database schema: Users, Incidents, EvidencePackages, MuleAccounts, SystemLogs, and TelemetryRecords.
services/api-gateway/Dockerfile	Production Node.js containerization recipe for multi-stage Docker builds.

Counterfeit Detection Service (services/counterfeit-detection/)

Code File Path	Exact Job & Technical Responsibility
services/counterfeit-detection/main.py	FastAPI microservice exposing /api/v1/counterfeit/verify endpoint. Implements RBI physical security verification rules: optical thread hue variance, intaglio edge sharpness, Mahatma Gandhi 3D watermark contrast, and tactile bleed lines relief.
services/counterfeit-detection/model_engine.py	Multi-spectral feature extractor and production ensemble engine. Extracts 181-dim robust percentile/gradient features and 142-dim color moments. Blends LightGBM, XGBoost, HistGradientBoosting, and ExtraTrees with calibrated class

	multipliers.
services/counterfeit-detection/models/inr_denomination_classifier.joblib	Serialized champion ensemble model achieving 95.06% validation accuracy across all 344 validation banknotes.
services/counterfeit-detection/models/model_metadata.json	Model performance metadata, per-class weights, feature dimensions, and calibration multipliers.
services/counterfeit-detection/models/genuine_security_profiles.json	Statistical feature moments (mean, std, min, max) of genuine RBI banknotes used for anomaly z-score calculation.
services/counterfeit-detection/test_logic.py	Unit test suite testing counterfeit detection rules and ₹10 banknote verification.
services/counterfeit-detection/test_main.py	HTTP endpoint integration test suite for FastAPI counterfeit service.

Scam Detection Service (services/scam-detection/)

Code File Path	Exact Job & Technical Responsibility
services/scam-detection/main.py	FastAPI microservice detecting digital arrest and extortion calls. Evaluates audio/speech transcripts using semantic pattern matching against known extortion templates (CBI arrest warrants, narcotics courier, customs parcels). Analyzes telecom metadata for CLI spoofing (HLR vs MSC mismatch).
services/scam-detection/test_logic.py	Unit tests for scam classification heuristics, extortion pattern detection, and spoofing scores.
services/scam-detection/test_main.py	API integration tests validating endpoint response structures.

Fraud Graph Service (services/fraud-graph/)

Code File Path	Exact Job & Technical Responsibility
services/fraud-graph/main.py	FastAPI microservice backed by Neo4j graph database. Traces multi-hop fund flows across UPI handles and bank accounts. Detects circular transactions, high-velocity layering, and executes automated account freeze orders.
services/fraud-graph/setup.cypher	Cypher script defining graph schema, node labels (:Account, :Citizen, :Mule, :Transaction), indexes, constraints, and baseline graph topology.

Threat Fusion Service (services/threat-fusion/)

Code File Path	Exact Job & Technical Responsibility
services/threat-fusion/main.py	FastAPI service calculating composite threat severity using Multi-Criteria Decision Analysis (MCDA). Aggregates signals from scam detection, counterfeit alerts, and banking graphs to compute national threat levels and automated LEA advisories.

Citizen Shield Service (services/citizen-shield/)

Code File Path	Exact Job & Technical Responsibility

services/citizen-shield/main.py	FastAPI multilingual citizen verification assistant (Hindi/English). Validates police notice authenticity, provides legal advisory, and prevents panic reactions during extortion attempts.

Evidence Generator Service (services/evidence-generator/)

Code File Path	Exact Job & Technical Responsibility
services/evidence-generator/main.py	FastAPI service generating cryptographically signed Digital Evidence Packages (DEPs). Produces forensic PDF dossiers containing incident timeline, audio waveforms, call logs, IP telemetry, and SHA-256 seal compliant with Section 65B of Indian Evidence Act.

4. Counterfeit Detection ML Model Benchmark (95.06% Accuracy)

The Counterfeit Vision subsystem employs a multi-spectral ensemble classifier trained on over 3,500 training images and rigorously validated against all 344 images in the official validation dataset across all 8 classes. The champion model combines LightGBM, XGBoost, HistGradientBoosting, and ExtraTrees with fine-tuned class calibration:

Class	Samples	Correct	Accuracy	Precision	Recall	F1-Score
₹100 (Lavender)	45	45	100.00%	1.0000	1.0000	1.0000
₹2000 (Magenta)	45	45	100.00%	0.9574	1.0000	0.9783
₹50 (Fluorescent Blue)	45	44	97.78%	0.9362	0.9778	0.9565
₹500 (Stone Grey)	45	44	97.78%	0.9167	0.9778	0.9462
₹20 (Greenish Yellow)	45	43	95.56%	0.9556	0.9556	0.9556
₹200 (Bright Orange)	44	41	93.18%	0.9111	0.9318	0.9213
₹10 (Chocolate Brown)	45	40	88.89%	1.0000	0.8889	0.9412
Background (Non-note)	30	25	83.33%	0.9259	0.8333	0.8772
Overall Dataset Total	**344**	**327**	**95.06%**	**0.9515**	**0.9506**	**0.9502**

5. How to Run the Complete System Locally

SentinelX is engineered for one-command local execution using Docker Compose, or modular execution via Node.js and Python:

Step 1 — Start Infrastructure & Microservices via Docker Compose:

Run the following command from the workspace root:

```
docker compose up -d
```

This launches PostgreSQL (5432), Neo4j (7474/7687), Redis (6379), Kafka (9092), API Gateway (8000), Scam Detection (8001), Fraud Graph (8002), Citizen Shield (8003), Threat Fusion (8004), Evidence Generator (8005), and Counterfeit Detection (8006).

Step 2 — Launch LEA Command Centre Dashboard:

From the apps/command-centre directory:

```
cd apps/command-centre
npm install
npm run dev
```

Open your browser to http://localhost:5173 to access the full operational Command Centre.

Step 3 — Run Automated Test Suites:

Verify system health and counterfeit model validation:

```
python -m unittest services/counterfeit-detection/test_logic.py
python -m unittest services/scam-detection/test_logic.py
python -m unittest agents/test_agent_mesh.py
```

6. Mobile APK Download & Phone Installation Guide

YES, you can now immediately download and install the SentinelX Citizen Mobile application APK file onto your Android phone! The standalone, production release APK has already been compiled, signed, and is stored directly on your computer.

WHERE IS THE APK LOCATED ON YOUR PC?

Primary Location (Workspace Root):
E:\Meet\Meet\Hackathon\SentinelX\SentinelX-CitizenMobile-v1.0.apk

Secondary Location (Android Gradle Build Folder):
E:\Meet\Meet\Hackathon\SentinelX\apps\citizen-mobile\android\app\build\outputs\apk\release\app-release.apk

File Size: 59.37 MB | **Package ID:** com.sentinelx.citizenmobile | **Version:** 1.0.0 (Release Build)

Step-by-Step Instructions to Install APK on Your Phone

Method 1: Direct USB File Transfer (Fastest & Easiest)

1. Connect your Android phone to your PC via a USB cable.
2. On your phone, swipe down the notification shade and select 'File Transfer / Android Auto' (MTP mode).
3. Open File Explorer on your Windows PC, copy the file 'SentinelX-CitizenMobile-v1.0.apk' from 'E:\Meet\Meet\Hackathon\SentinelX\'.
4. Paste the APK into your phone's 'Downloads' or 'Internal Storage' folder.
5. Open the 'Files' or 'File Manager' app on your phone, tap 'SentinelX-CitizenMobile-v1.0.apk', and tap 'Install'.

Method 2: Wi-Fi Wireless Download via Python Local Server

1. Make sure your PC and Android phone are connected to the SAME Wi-Fi network.

2. Open a command prompt or terminal in the SentinelX workspace and run:
   python -m http.server 8080
3. Find your PC's local IP address (e.g., 192.168.1.15) by running 'ipconfig' in command prompt.
4. On your Android phone, open Chrome or any browser and type:
   http://<YOUR_PC_IP>:8080/SentinelX-CitizenMobile-v1.0.apk
5. The browser will instantly download the APK file directly to your phone. Tap to install!

Method 3: Cloud / Messaging Share (Google Drive / WhatsApp Web)

1. Open WhatsApp Web or Google Drive on your computer.
2. Send or upload 'SentinelX-CitizenMobile-v1.0.apk' to yourself or save it in your Google Drive.
3. On your phone, open WhatsApp or Google Drive, tap the file, and proceed with the installation.

Method 4: Direct Developer Install via ADB (Android Debug Bridge)

If you have USB Debugging enabled on your phone and the Android SDK / ADB installed:

1. Connect phone via USB.
2. Run in terminal:
   adb install E:\Meet\Meet\Hackathon\SentinelX\SentinelX-CitizenMobile-v1.0.apk
3. ADB will stream and install the app onto your phone within 5 seconds with 'Success'.

> ⚠ **Note on Android Permissions ('Install Unknown Apps'):**
> When installing an APK directly for the first time, Android will prompt: 'For your security, your phone is not allowed to install unknown apps from this source'. Simply tap 'Settings', toggle ON 'Allow from this source', and tap 'Install'. This is standard for all sideloaded hackathon release APKs.